The Lonely Landmark

A lightweight pipeline for automatic object removal in tourist photography

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Outline

- Problem statement
- State of the art
- Proposed method
- Dataset
- Experimental setup
- Model evaluation
- Conclusions

Problem Statement

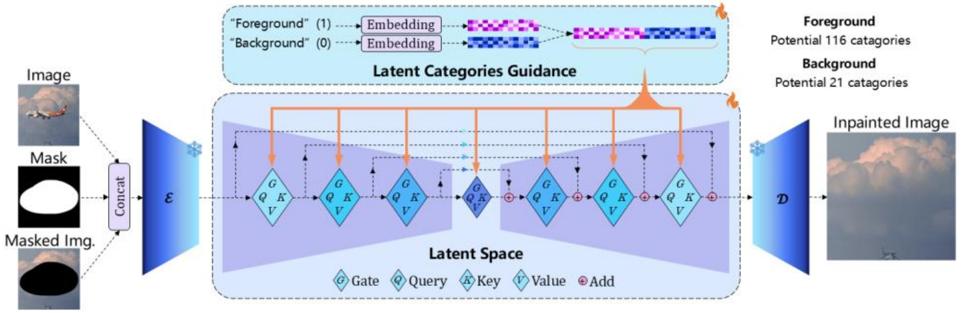
- Tourist photos are crowded with strangers
- Manual removal needs costly tools or paid cloud APIs
- Existing tools: heavy models, manual masks, manual prompts
- Goal: cheap, fast, fully automatic removal — one click

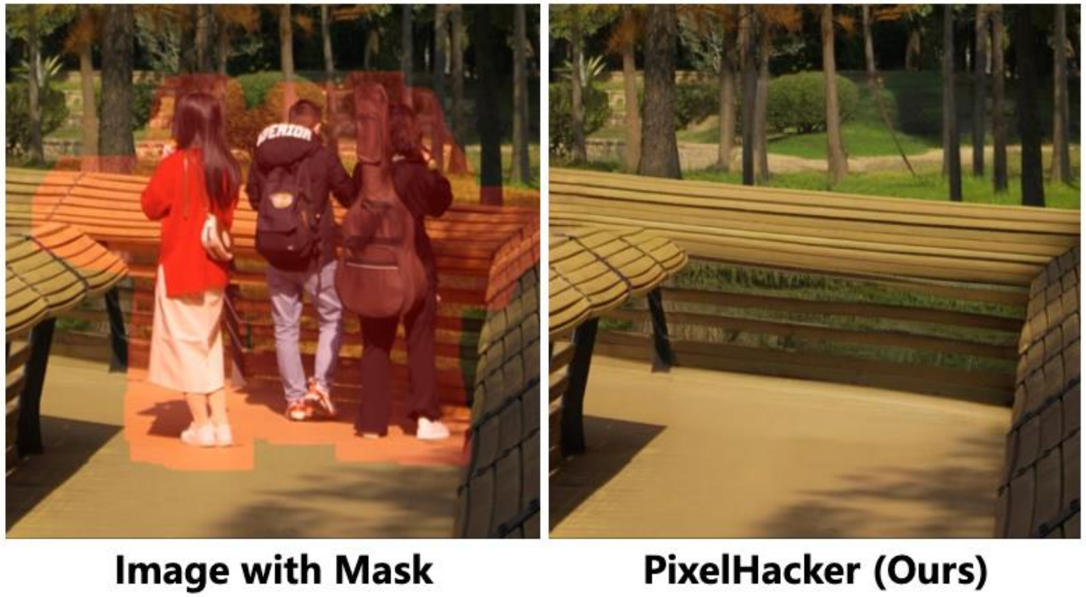
Input (320, 250)

Improved
18.7s

State of the Art

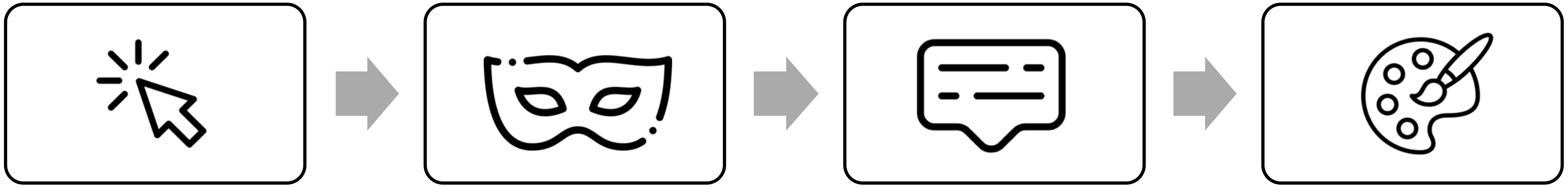
- **PixelHacker** - Xu et al., HUST & VIVO AI Lab (2025)
- Trains a 0.8B diffusion model from scratch for inpainting
- Novel Latent Categories Guidance for structural & semantic consistency
- SOTA on Places2, CelebA-HQ, FFHQ
- Needs scene-specific fine-tuning, a separate checkpoint per dataset





Proposed Method

Click → SAM mask → BLIP-2 caption → RePaint inpaint → evaluation



Improved Pipeline - 8 Refinements

Vanilla RePaint + 8 targeted refinements — recovering quality on a small SD1.5 backbone

1 · Sampling-process fixes (how the diffusion loop runs)

1. **Coherent noise**
2. **Selective resampling**
3. **Guidance ramp**
4. **Progressive mask**

2 · Seam hiding (post-sampling)

1. **Laplacian blending**
2. **SDEdit refinement**

3 · Hygiene

1. **Deterministic VAE**
2. **Negative prompt**

Refinements - Sampling Process

Compensating for reduced model capacity inside the diffusion loop

1. **Coherent noise** — one shared noise tensor keeps the fill and its surroundings evolving together
2. **Selective resampling** — jump-back resampling restricted to mid-denoising, where harmonization helps
3. **Guidance ramp** — CFG rises 5 → 9 across steps: flexible early, prompt-locked late
4. **Progressive mask** — mask dilated & blurred while noisy, then tightened as the image sharpens

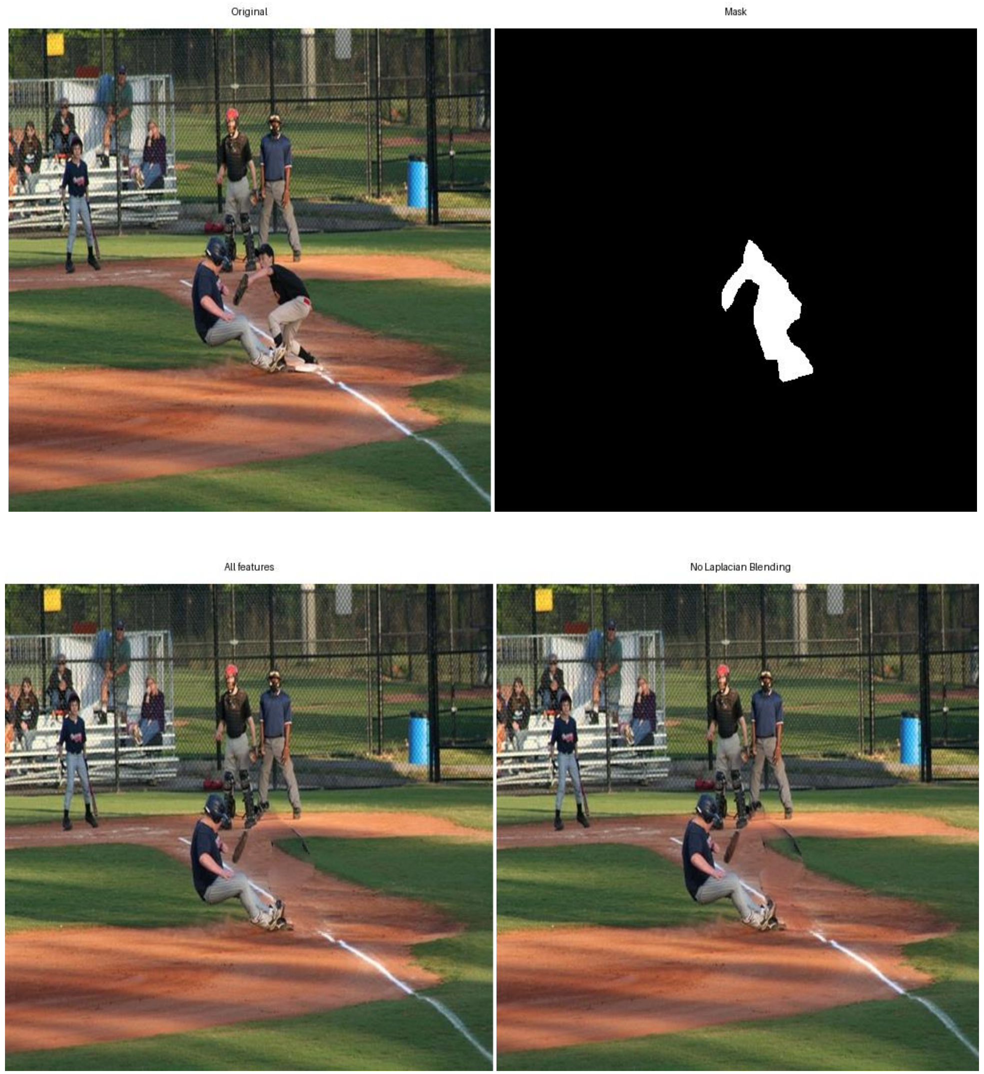
Refinements - Seam Hiding & Hygiene

Seam hiding

1. **Laplacian pyramid blending** — blends fill into the original per frequency band, dissolving the seam
2. **SDEdit refinement pass** — adds light noise & re-denoises the whole image to smooth artifacts

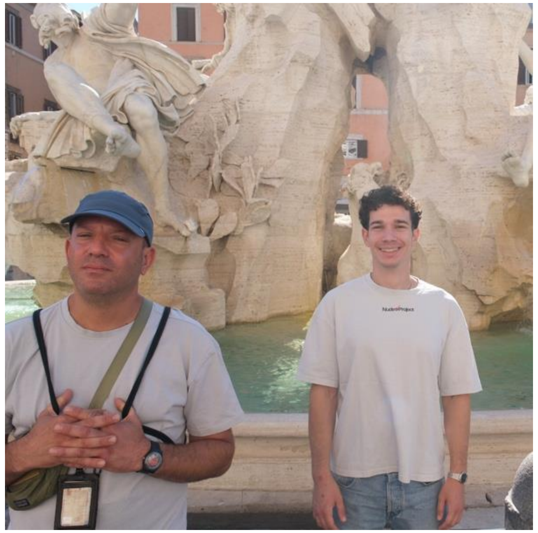
Hygiene

1. **Deterministic VAE encoding** — encodes via mode() — removes a noise source, reproducible runs
2. **Default negative prompt** — steers away from people, faces, seams, lighting mismatch



Dataset

- COCO 2017 val — instance masks as ground-truth targets
- Filtered: medium objects, person category
- Our own Rome photos — qualitative, real-world demo



Experimental Setup

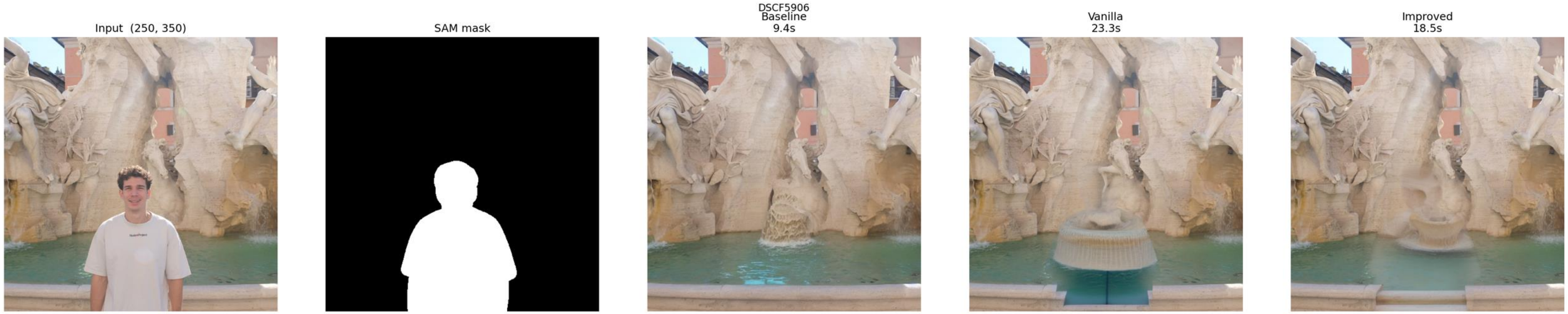
- Three pipelines: baseline / vanilla / improved
- Vanilla & improved share identical SD1.5 weights
- 512px, 50 steps, seeds 42 / 123 / 7
- Metrics: LPIPS, mask-LPIPS, CLIP score

Model Evaluation

- Baseline vs vanilla vs improved
- Lower LPIPS / mask-LPIPS = closer to ground truth
- Higher CLIP = better prompt match

	LPIPS ↓	Masked LPIPS ↓	CLIP ↑	Time in seconds
Baseline	0.15976317	0.353364367	0.2404644	8.157733333
Vanilla	0.09943017	0.338231	0.2425251	22.05073333
Improved	0.0905175	0.308282033	0.2425251	17.2656

Model Evaluation



Model Evaluation

- Disable one refinement at a time
- 30-sample subset, single seed
- Shows which refinement matters most

Conclusions

- Fully automatic, one-click removal is feasible
- Targeted refinements recover quality on a small backbone
- Biggest gains: Coherent noise, Selective resampling.
- Speed: our improved pipeline was always faster than the vanilla pipeline.
- Future: video, higher resolution, faster sampling.

References

- Rombach et al. — High-Resolution Image Synthesis with Latent Diffusion Models
- Lugmayr et al. — RePaint: Inpainting using DDPMs